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Social Media and Online Communities Report Analysis

Student Names& ID:-

- Essa Alyafei (s5087744)
- Hassan Jaafar (s5087726) • Khalid Al-Humaidi (s5087505)
- Mollie Holdaway (s5622642)
- Osman Choudhury (s5646583)
- Paulius Laukaitis (s5608606)
- Rahaf Al-Malki (s5087725)

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Introduction

1.1: Overview of Social Media and Online Communities

Online communities, which were first introduced in the 1970s, have evolved into complex social networks, transforming human interaction and business landscapes(Mandal, 2023). To define, Social media platforms are widespread, connecting individuals worldwide. They have transformed people's interactions by moving in-person communication to a digital platform. This journey has made it easier for people with similar interests and passions to connect, collaborate, and share ideas, thereby strengthening relationships(Jones, 2023).

Social media platforms have transformed communication, entertainment, and business, with billions of users worldwide incorporating them into their daily lives. Cooperation and communication are essential for success in today's business environment because they improve accessibility and allow people to stay connected regardless of distance. Second, improved collaboration in a variety of areas, including workplace productivity and global networking, such as the installation of different platforms that allow for quick discussions and decision-making. Finally, it improved transparency by enabling individuals and businesses to share updates instantly (Okonkwo and Awad, 2023). While also social media played a significant role in entertainment as social media has empowered entertainment brands to significantly improve the effectiveness of their advertising strategies. They can now promote their projects directly to the right audience on a large scale, which results in higher conversion rates and engagement (Brisset, 2023). Businesses have increased awareness by developing social media and online communities, which establish and shape brands in terms of languages and messaging, sales and marketing, and content distribution (Hiscox, 2023).

We have selected four different software Businesses for the chosen segment to focus on our analysis as they are primary companies in this rapidly changing industry each platform has its own set of features and targets different demographics of users, adding to the social media ecosystem’s diversity. Below are the selected software businesses selected by the group:

1. **Snapchat:** users can send and receive "self-destructing" photos and videos.
2. **Instagram:** a photo-sharing app, has grown to become a major hub for influencer marketing and online shopping
3. **TikTok:** a revolution in brief video content through its strong, AI-powered suggestion system
4. **Twitter(x):** continues to be a center for up-to-the-minute news, talks, and ways to make money online.

The global social media industry is upwards spiraling with customer interactions, advertising achievements touching the sky, and even some new proposals like AI technology and blockchain-based monetization arising. Nonetheless, such networks are caught up in a myriad of key issues such as political scrutiny, data breach scandals, fake news outbursts, and rivalry from the cutting edge outlets.

1.2: Purpose and objective of the report

This is a report that aims at a comprehensive analysis of the social media software business segment, which is a new way to make fun, everyday life. It includes such applications as Snapchat, TikTok, Instagram, and Twitter (X). The main goals of this analysis consist the following aspects:

- Determining the differences and market positions of these platforms.
- Defining their business models and the technology and software that are behind their success.
- Testing the adoption of software strategies in their business goal agenda.
- Ratings of professionals, legalities, ethical, and sustainability considerations within the segment.

1.3: Research Questions and Report Structure

In order to meet these goals, the present report is aimed at exploring the main research questions by exploring the unique traits of Snapchat, TikTok, Instagram and Twitter (X) in today's market, the business models and software/IT technologies that make their growth and competitive advantage possible, the alignment of the software strategies with commerce objectives as well as the legal, ethical, and sustainability issues.

The report is organized as follows:

Section 2

provides a detailed analysis of the unique features of the mentioned platforms, including their main qualities, market trends, hurdles, and opportunities.

Section 3

deals with the business models of the platforms and software/IT technologies and focuses on their effect on business strategies, organizational structures, and product development.

Section 4

reviews the partnership between business and IT strategies, considering the role of new technologies.

Section 5

covers the examination of the legal, professional, ethical, and sustainability issues of these companies.

Section 6

the summary of the findings and gives some suggestions for the growth and innovation of the companies in the social media industry. The main aim of this report is to give new insights into the business segment of social media, outline the strategies employed by the leaders, and in so doing, offer practical suggestions to the business owners.

1. Snapchat

Snap Inc.'s Snapchat, launched in 2011, rapidly distinguished itself within the crowded social media and communication software industry. Its pioneering approach to ephemeral messaging—content designed to disappear—and its early, widespread adoption and popularisation of consumer-facing augmented reality (AR) established it as an innovative force, particularly appealing to younger demographics (Snap Inc, 2024). This report undertakes a critical analysis of Snapchat, examining its unique position within the software business segment. It explores its changing business models, the advanced IT and software technologies that support its platform, the strategic alignment of its technological capabilities with business goals, and its complex engagement with sustainability, ethical, legal, and professional issues. The goal is to analyse Snapchat's operational dynamics, competition management, and probable future in a fast-changing, technologically advanced digital environment.

2.1: Unique Aspects of Snapchat in the Software Business Segment

Snapchat operates in the visual-first social media segment, known for high engagement, rapid innovation, and intense competition for younger demographics (Statista, 2024a). Its ephemeral content model influences communication trends and fosters demand for authentic digital interactions (van Dijck, 2013).

Key characteristics define Snapchat: ephemeral content (disappearing "Snaps") encourages spontaneous, less curated sharing and offers a perception of privacy (Bayer et al., 2016). It leads in consumer AR with Lenses, extending into AR-powered commerce and branded content (Snap Inc., 2025; Roy, 2019). The platform strongly appeals to users aged 13-34, especially Gen Z (Statista, 2024; Dixon, 2024), though this presents challenges in broadening its user base. Its camera-first interface prioritises immediate visual creation (Snap Inc., n.d. a), and continuous innovation is shown through features like Stories and Spotlight (Wagner, 2015).

The competitive landscape is fierce. Meta (Instagram) replicates features and competes for AR engagement and ad revenue (Isaac, 2016). TikTok dominates shortform video (Herrman, 2019). Figure 1 illustrates Daily Active User (DAU) trends, showing Facebook's larger user base against Snap's substantial, consistent numbers.

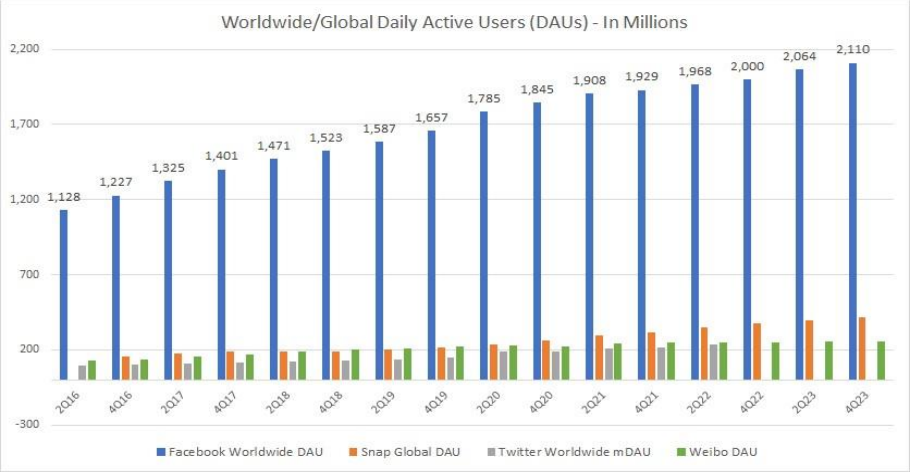

Figure 1Worldwide/Global Daily Active Users (DAUS) - In Millions from Q2 2016 to Q4 2023 for Facebook, Snap, Twitter, and Weibo. (Source: Frontman (2024))

Market drivers include demand for engaging content, mobile and AR/AI advancements, and innovative ad solutions. The network effect is also critical (Katz and Shapiro, 1985).

Snapchat faces challenges: diversifying monetisation beyond advertising, despite Snapchat+ growth (Snap Inc., 2024d; Iqbal, 2017); feature replication by larger rivals; user churn and expanding demographic reach (Carman, 2018); and competing with advanced recommendation algorithms. Economic conditions also impact ad revenue (Vanian, 2023). Opportunities lie in AR e-commerce, enterprise AR (Snap Inc., 2023a), Snapchat+ expansion, and growth in untapped international markets.

2.2:- Business Models and Related Software/IT Technologies

Snapchat's primary business model is advertising-driven. Most revenue stems from innovative ad formats, including full-screen vertical video Snap Ads, image ads, and interactive ads seamlessly integrated between Stories, Discover content, or Spotlight videos. Sponsored AR Lenses allow brands to create custom, interactive AR experiences that users engage with and share, offering high engagement. Sponsored Geofilters provide location-based overlays for specific events or locales (Snap Inc, 2024; Snap Inc., n.d. b). A strategically important secondary revenue stream is Snapchat+, a subscription service (launched June 2022) offering subscribers exclusive, experimental, and pre-release features, diversifying revenue beyond advertising (Snap Inc., 2024d). Revenue from hardware sales, specifically its Spectacles AR glasses, remains a minor contributor, generally perceived as long-term R&D for future AR interaction rather than a current profit centre (Snap Inc., 2023b).

Snapchat's robust functionality and unique user experience are underpinned by a sophisticated, evolving software and IT stack. It heavily relies on third-party cloud infrastructure, primarily Google Cloud Platform (GCP) and Amazon Web Services (AWS). This provides scalability, global reach, robust storage for vast user-generated content (Snaps, Memories), powerful processing for features like AR rendering, and efficient content delivery (Snap Inc, 2024; Lawler, 2017). Artificial intelligence (AI) and

machine learning (ML) algorithms are integral to many core functions. These technologies optimise image/video processing (Roth, 2024), support content moderation, provide exact ad targeting, power material suggestions (Discover, Spotlight), and enable face recognition for AR Lenses. When creating native ios and Android apps that are optimised for a seamless user experience, mobile-first development is essential. Its Augmented Reality Development Kit, Lens Studio, represents a significant technical distinction. It promotes a thriving ecosystem of sponsored and user-generated AR experiences by enabling a worldwide community to create and share AR Lenses (Snap Inc., 2024b).

While specifics of Snapchat's proprietary backend architecture are not fully public, similar large-scale platforms typically employ a sophisticated mix of database technologies. This often includes Nosql databases, such as Amazon Dynamodb (conceptualised in Figure 2), to efficiently handle massive volumes of unstructured user data and ensure high availability, fault tolerance, and horizontal scalability. Figure 2 illustrates a conceptual backend architecture for a large social application. Such an architecture showcases components like a highly scalable NoSQL database, a configuration management system for dynamic application settings, API servers (like an xDS server for traffic management), an API Gateway for external traffic, and loosely coupled microservices for core functionalities (Friends graph, User authentication, App integrations, Settings). This microservices-based style allows modularity, independent service scalability, and efficient development cycles.

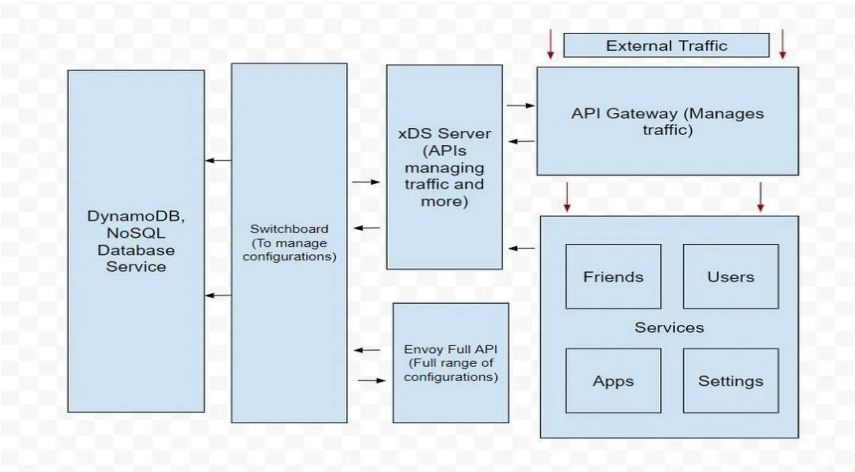

Figure 2Conceptual Backend Architecture for a Large-Scale Social Application, illustrating components like Nosql databases, configuration management, and microservices. (Source: Liu (2022))

Real-time communication protocols (e.g., WebRTC, XMPP) are essential for core interactive features like instant messaging, voice/video chat, and Snap Map live updates. The overall technology stack for building a Snapchat-like application, as

depicted in Figure 3, often encompasses diverse development languages (e.g., Python, Swift, Java/Kotlin), robust hosting solutions (e.g., Google Compute Engine, aligning with Snapchat's GCP use), various front-end frameworks, scalable back-end services (e.g., Google App Engine), and a suite of third-party tools for marketing, sales, and advanced user analytics.

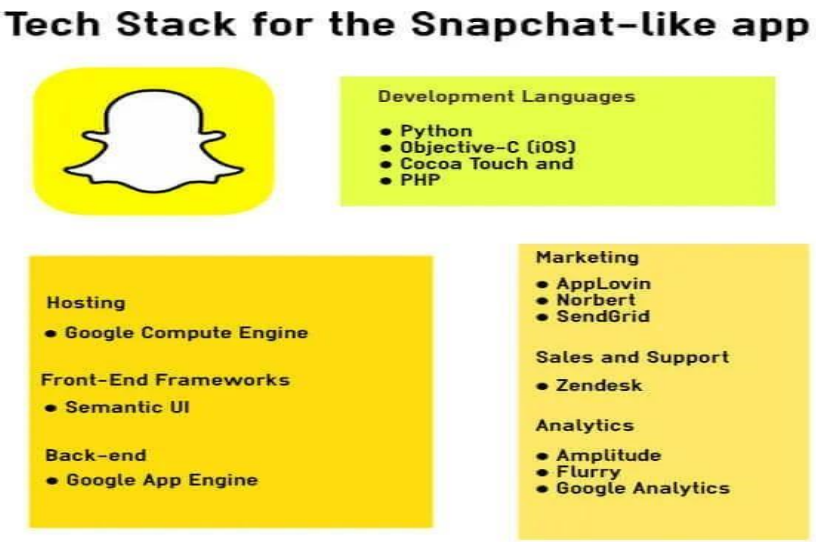

Figure 3Example Tech Stack for a Snapchat-like Application, detailing potential development languages, hosting, backend, and other operational tools. (Source: Denys (2023))

Figures 2 and 3 show how IT and supporting software affect Snapchat's organisational structure, product development, and business strategy. AI and ML are the driving forces behind the company's advertising-centric approach, which increases user engagement and advertiser ROI via precise ad targeting and individualised content. Cloud computing's scalability allows for global user growth and peak traffic load management without necessitating large upfront investments in physical infrastructure. Its cutting-edge AR technology, which enables the creation of immersive brand experiences and unique advertising formats that command premium pricing, is one of its primary differentiators (Snap Inc., n.d. b). In terms of organisational structure, the necessity for quick innovation in AR and AI calls for cross-functional, agile teams that combine product, design, engineering, and data science capabilities to speed up development. Dedicated teams focus on platform integrity, user safety, and content moderation, using sophisticated AI tools and human reviewers to enforce guidelines (Snap Inc., 2025). Snapchat uses an agile approach to product development, which is characterised by rapid incremental updates, A/B testing, and a data-driven refining process. DevOps procedures and CI/CD pipelines assist this, enabling quick innovation and reaction to user input and market trends.

2.3:-Business and Software/IT Alignment Strategies

Snapchat's software and IT strategies are intrinsically linked to its mission: "to empower people to express themselves, live in the moment, learn about the world, and have fun together" (Snap Inc., n.d. a). The camera-first design, deep AR integration (including Lens Studio), ephemeral content, and real-time sharing capabilities directly support this vision. Content platforms like Discover and Spotlight, powered by AI, facilitate learning and discovery. IT strategies that enhance user engagement directly bolster the advertising revenue model.

New software technologies are pivotal for Snapchat's evolution. Advancements in AI/ML, particularly generative AI, are explored for novel Lenses, creative tools (e.g., "My AI" chatbot), and more dynamic ad content (Roth, 2023). AI is also vital for scaling content moderation. Cloud computing remains essential for global scalability and managing computational demands, with edge computing being explored to reduce AR latency. While not at the core, Snap Inc. observes blockchain and Web3 for potential future applications like creator monetisation (Newton, 2022). Investment in sophisticated AR hardware (Spectacles) aims to deepen immersion, though mass adoption is a long-term goal (Snap Inc., 2023b; Heath, 2021).

Aligning these advanced software developments with business strategies presents challenges. The resource-intensive nature of R&D in areas like AR requires careful management to ensure ROI against well-funded competitors. Balancing a seamless user experience with ad density is crucial to avoid alienating users. The ethical implications of increasingly sophisticated AI, such as algorithmic bias or misuse of generative AI, require proactive governance and trust-building measures (The Editors, 2025). Successfully integrating hardware like Spectacles into the ecosystem and driving mass adoption has proven difficult (Snap Inc, 2024). Finally, quantifying the ROI of emerging tech investments complicates strategic resource allocation.

2.4:- Professional, Legal, Ethical, and Sustainability Considerations

Snap Inc. aims for the greatest professional standards in platform governance, data security, and software engineering, including content moderation systems (Snap Inc., 2025). Still, a significant operational issue remains the effective and dependable management of vast amounts of user-generated content.

Legally, Snapchat navigates a complex global landscape. Data protection laws like GDPR (EU) and CCPA/CPRA (California) dictate how user data is handled, requiring consent, providing user rights (access, deletion), and mandating breach notifications (Information Commissioner's Office, n.d.; State of California Department of Justice,

n.d.). Given its large youth audience, child online safety legislation (e.g., COPPA in the US, UK Online Safety Act, EU Digital Services Act) imposes stringent obligations regarding age verification, protection from harmful content, and feature design (Federal Trade Commission, n.d.; Ofcom, 2023; European Commission, n.d.).
Intellectual property protection is also a key legal concern.

Ethical considerations are multifaceted. User well-being and mental health, particularly for younger users, are subject to scrutiny regarding potential impacts of social media use, AR filters, and ephemeral content (Royal Society for Public Health, 2017; Twenge, 2019). Like other platforms, Snapchat has faced lawsuits alleging addictive product designs harmful to minors (Allyn, 2023). Effective content moderation across diverse cultural contexts is an ongoing ethical and operational hurdle. Data privacy, especially concerning ad targeting with user data, remains a prominent ethical debate (Zuboff, 2019). The potential for **algorithmic bias** in AI systems for content curation or AR Lenses requires continuous monitoring and mitigation (Noble, 2018).

Table 1Key Legal Frameworks and Snapchat's Compliance Approach & Challenges (Source: Self-made)

Legal Framework (Jurisdiction)	**Key Provisions/Focus**	**Snapchat's General Compliance Approach/Features**	**Key Challenges for Snapchat**
GDPR (General Data Protection Regulation) (EU/EEA)	Data protection and privacy; user consent; data subject rights (access, rectification, erasure, portability); data breach notifications; restrictions on processing children's data.	Privacy Centre; detailed Privacy Policy; tools for users to download or delete their data; age-gating mechanisms; consent flows for data processing (e.g., personalised ads); appointment of a Data Protection Officer (DPO); emphasis on data minimisation through ephemeral features.	Ensuring valid and granular consent across all features; robust age verification for young users; managing cross-border data transfers; the complexity of responding to Data Subject Access Requests (DSARS) at scale; keeping pace with evolving interpretations from Data Protection Authorities (DPAS).

CCPA (California Consumer Privacy Act) / CPRA (California Privacy Rights Act) (California, USA)	Consumer rights to know, delete, and opt out of the sale/sharing of personal information; right to limit the use of sensitive personal information; nondiscrimination for exercising rights; data security requirements.	Provides mechanisms for users to access and delete their data; "Do Not Sell or Share My Personal Information" options in settings; updates to privacy policy reflecting CCPA/CPRA rights; processes to honour opt-out requests.	Defining "sale" and "sharing" in the context of complex ad tech ecosystems, implementing effective opt-outs across various data flows, robustly verifying consumer identity for requests, managing sensitive personal information as defined by CPRA, and staying updated with evolving regulations and enforcement.
COPPA Privacy (Children's Online Protection (USA)	Parental consent is required for collecting personal information from children under 13; clear privacy policies regarding data collection from children are required.	Terms of Service state that users must be 13+, age-gating at sign-up, aims to prevent under-13s from creating accounts, provides reporting mechanisms for underage accounts, and states that it does not knowingly collect personal information from children under 13 without parental consent.	Difficulty of implementing foolproof age verification (reliance on selfattestation); children misrepresenting their age to access the platform; ensuring ad targeting and data practices remain compliant if an under-13 user bypasses age gates; potential scrutiny due to its large youth user base.

UK Online Safety Act (United Kingdom)	Duty of care for platforms to protect users (especially children) from illegal and harmful content; risk assessments; transparency reporting; robust age verification for certain content; tackling CSEA and terrorist content.	Investments in safety tools (e.g., Family Centre, reporting); AI and human content moderation; cooperation with law enforcement; development of age assurance technologies; publication of transparency reports on content moderation and enforcement actions.	Defining and consistently moderating "legal but harmful" content; implementing effective and privacy-preserving age verification at scale; balancing user safety with freedom of expression and privacy (especially with encrypted messages); resource intensity of proactive content
			moderation and risk assessments.
DSA (Digital Services Act) (EU)	Harmonised rules for online intermediaries; combating illegal content, goods, and services; increased transparency (ads, recommender systems); risk management for Very Large Online Platforms (VLOPs); user redress mechanisms; protection of minors.	(As a likely VLOP) Enhanced content moderation systems; increased transparency regarding advertising and content recommendation algorithms; user reporting and appeal mechanisms; conducting regular risk assessments; specific measures to protect minors online; appointing legal representatives in the EU.	Meeting extensive obligations for VLOPS; achieving true transparency in complex algorithmic systems without revealing proprietary information; managing diverse types of illegal content across 27 EU member states; ensuring recommender systems are not manipulative or discriminatory; significant resource allocation for compliance.

Sustainability initiatives, as described in CitizenSnap reports (Snap Inc., 2023c), focus on social and environmental factors (DEI, civic engagement, mental health resources, and "Safety by Design" principles) as well as environmental impacts (carbon footprint reduction, energy efficiency with cloud providers like Google Cloud, n.d., and Amazon Web Services, n.d.). However, like other internet giants, Snap often faces scrutiny regarding the true scope and effectiveness of these programs, given their impact on the global platform (Sustainalytics, 2023). However, like other internet behemoths, Snap is often questioned about the real scope and efficacy of these programs in light of their influence on the worldwide platform (Sustainalytics, 2023).

Through its advances in consumer AR and ephemeral content, Snapchat has established a distinct market position. Its primary advantages are its agility and high attraction to younger populations. It does, however, function in a highly competitive market (Figure 1), where it faces difficulties in diversifying monetisation beyond advertising, thwarting feature copying by more established competitors, and negotiating a convoluted web of moral and legal requirements. The alignment of Snapchat's IT strategy with its business objectives is evident, particularly in its AR and AI focus, supported by scalable backend systems (Figures 2, 3). Yet, translating this technological leadership into consistent, profitable growth remains an ongoing challenge (Snap Inc, 2024). Snapchat's future success will depend on its ability to deliver continuous, hard-to-replicate innovation, expand revenue streams (with Snapchat+ showing promise), potentially broaden its user base without alienating core users, and proactively manage the significant professional, legal, ethical, and sustainability considerations inherent in its global operations.

3.TikTok:

AI driven content discovery model is what has made TikTok to stand out as the second emerging force in the digital space. When TikTok was launched globally in 2017 by ByteDance, it formed the opposite of a conventional social media, basing algorithmic engagement over personal network. Gen Z and Millennial users were seduced by this and completely changed global culture, advertising, and music promotion. Creators can reach viral on the platform with or without follower count. This report, critically examines the business model, technological foundation, technological form, legal, ethical and sustainability aspects of TikTok as the company rapidly evolves to become the central player in the global digital economy.

1. Unique Aspects of TikTok as a Business

1.1 Unique Features: Short-Form Video and Algorithm-Driven Discovery

Short form video content paired with both A.I. recommendation is what makes Tiktok. In the beginning, TikTok was 15 second videos, but then the longer 3 minute formats added to video capacity but kept the briefness. For You page (FYP) uses machine learning to personalize content feeds that anyone can go viral no matter, how much followers (see Figure 1) [TikTok, 2023].

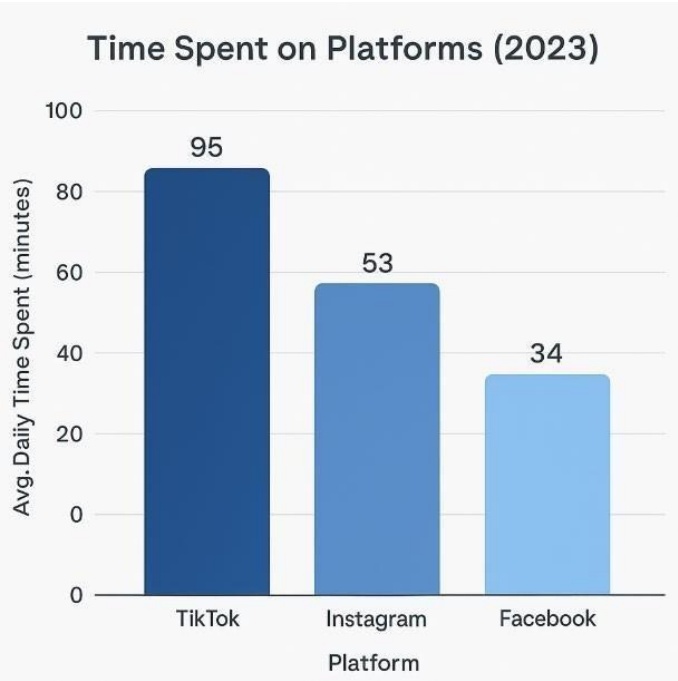

Figure 1 TikTok's Unique Content Discovery Model Source: TikTok Business Blog (2023)

1.2 Study of Market Trends and User Engagement Growth

Since its launch globally in 2017, TikTok grew exponentially and now has over 1,2 billion monthly active users in 2023. It has very high engagement with average users spending 95 mins every day as compared to Facebook and Instagram. Such strong engagement is primarily favorably attributed to TikTok's algorithmic precision and its youth oriented content delivery (see Table 1) [Statista, 2023]. *Table 1 User Engagement Comparison (2023) Source: Statista (2023)*

Platform	Avg. Daily Time Spent	Monthly Active Users	Dominant Age Group
TikTok	95 minutes	1.2 billion	16–24 years
Instagram	53 minutes	2 billion	18–34 years
Facebook	34 minutes	2.91 billion	25–44 years

1.3. Overview of TikTok's regulatory and content moderation challenges

TikTok faces significant regulatory and content moderation challenges globally. Concerns include data privacy, national security, and the platform's influence on youth. For instance, the European Union has urged TikTok to enhance its content moderation to comply with the Digital Services Act. U.S. policymakers are also nervous about data flows to China through TikTok's parent company, ByteDance, and will continue to be so until they can be addressed. These challenges require region specific solutions to compliance such as Project Texas in the U.S. and regional data centers in the EU [Reuters 2023a].

1.4 Opportunities: AI-Driven Personalization and Expansion into E-Commerce

Social commerce is now the future of growth for TikTok, leveraging their personalized content model to turn into an e-commerce engine. TikTok Shop and live shopping features bring projected gross merchandise value (GMV) of the TikTok business to $20bln in 2023 (see Figure 2) [Reuters, 2023b].

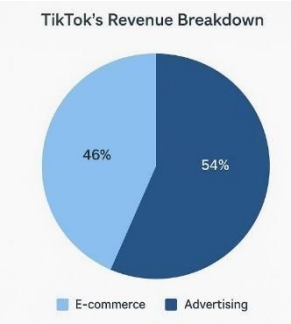

Figure 2 TikTok's Revenue Breakdown Source: Business of Apps (2023)

2. Business Models of TikTok

2.1 Revenue Model: Advertising, Live-Stream Gifting, and E-Commerce

The revenue model of TikTok blends user engagement, creator monetization as well as in app commerce. This model is based on three main streams of its core: advertising, live-stream gifting, e-commerce through TikTok Shop.

Mostly it is advertising revenue with advertising formats like In-Feed Ads, TopView Ads, Branded Hashtag Challenges and Branded Effects (TikTok, 2023a). What is special about TikTok is its AI powered targeting since to deliver ad, TikTok optimizes based on what user watch time, likes and shares. As for Gen Z and millennials, personal behavioral marketing has been very effective because it's relevant and somewhat non-intrusive.

The gifting lives model serves as a way for the users to buy in app coins and send virtual gifts to the creators during their live-stream. Using these gifts, TikTok collects a commission, establishing a peer to peer revenue system, that supports and stimulating the producer economy and producing content (Business Insider, 2023).

TikTok also created e-commerce on the platform with the TikTok Shop. It enables brands and influencers to show, sell the products through live videos and shoppable content and it brings product discovery, entertainment and purchasing together in a seamless way (Reuters, 2023). TikTok Shop has been extremely successful in Southeast Asia and is growing to Western markets with high conversion rates for brands (Planly 2023).

Table 2 TikTok Revenue Streams Overview

(Source: Compiled from TikTok Business Help Center and Business of Apps, 2023)

Revenue Stream	Description	Monetization Method

Advertising	In-feed, TopView, branded content	CPM, CPC, brand partnerships
Live-Stream Gifting	User donations during creator livestreams	In-app coin purchases
E-commerce (TikTok Shop)	In-app product sales, live promotions	Commission-based model

2.2 Key Technologies: AI and Deep Learning in Content Curation

Especially, artificial intelligence infrastructure of TikTok drives content recommendation, moderation and trend detection. As shown in Figure 3, recently, there is a Feed Page (Zhang et al., 2021), which also relies on machine learning model to learn user behavior and predict what videos interest them, based on a combination of things like video watch time, likes, replays and shares. Data from context such as location, device, audio etc is also considered.

Collaborative filtering, reinforcement learning and vision models are driven to predict these results. The more users engage with system, the more it gets better at generating in real time refined recommendations, with resulting higher retention and longer engagement (Forbes Tech Council, 2021).

According to the recommendations, TikTok also employs deep learning, such as 'CNNs' for video image analysis and 'NLP techniques' to extract the captions, hashtags and keywords to automatically classify content and detect trends at scale.

Moderation and discovery without over manual review (TikTok, 2023b) is possible because of this process.

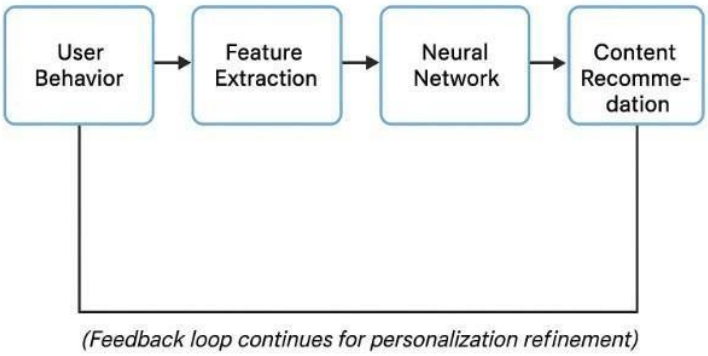

Figure 3 Simplified TikTok AI Pipeline (Source: Zhang et al., 2021)

2.3 Impact on Business Strategy, Organization, and Product Development

The business model of TikTok is directly shaping the strategic vision, organizational structure and product development of the company. The entertainment e commerce convergence model in the company is trying to marry e commerce and entertainment

layers into one seamless experience. TikTok Shop and in-video product tagging allow creators to simultaneously entertain and sell (Diginyze, 2023).

TikTok's modular, agile structure and the teams with focused responsibilities of Creator Partnerships, E-Commerce Ads and Trust & Safety, all support the parent company's claims. The decentralization features enable rapid feature deployments as well as regional compliance (e.g., GDPR in the EU, Project Texas in the US) (TikTok Transparency Center, 2023).

It is very iterative and data driven product development. Real-time dashboards and A/B testing guide feature rollouts such as Stories and extended videos. However, towards regulatory scrutiny and transparency results are made in efforts to implement explainable AI (XAI) (Forbes, 2021).

Figure 4 Evolution of TikTok's Business Model (Source: Compiled from CB Insights, 2023)

3.1 AI-Powered Engagement and Its Impact on User Retention

The AI recommendation engine at the center of TikTok's software strategy is the source of personalized content delivered to users through the platform's 'For You Page' (FYP). Unlike traditional social networks that are dependent on the relations between users expressed with the user following graph, TikTok reduces user friction by taking user behaviour into account, i.e. relying on video behavioural data such as watch duration, likes, shares, and swipe patterns. This has been a huge factor in TikTok reaching such high user retention as studies show TikTok users spend more time on TikTok than any other large social platform (Zhang et al., 2021).

Most of our engagement loop is powered over time by advanced machine learning models such as collaborative filtering and reinforcement learning which gradually improves the content ideas that are shown to you in real time. With continuous updating of the system's knowledge of user preferences, it increases the ad exposure and product discovery. From a business angle, a funding machine is created, driven by advertising impressions and e-commerce conversion, but the business does not dictate the exposure of how the creator economy should work, the relevance— not the follower count—is the driver.

3.2 Challenges in Balancing Algorithm Transparency and Regulatory Concerns

TikTok is growing in regulatory scrutiny despite the success of its technology. Users complain of ignorance about why their content gets viewed, and that this is too much of an algorithm and a lack of privacy. Frameworks such as GDPR and the Digital Services Act are in compliance demanding thereof from regulatory bodies, and that is especially so in the EU and U.S.

To this, TikTok has offered Transparency Reports, Transparency and Accountability Center, and some user controls. That could however put the platform open to exploitation if it disclosed too much about its algorithms. The choice to invest in Explainable AI (XAI) is a compromise for TikTok to blend with the regulatory demands and user trust, that is, both being essential for long term sustainability (TikTok, 2023).

4. TikTok: Professional, Legal, Ethical, and Sustainability Considerations

TikTok is expanding globally, and with the growth, it is attracting more scrutiny in four key areas: ethics of using AI, legal compliance, sustainability, and professional responsibility. Now, these dimensions are at the core of how digital platforms will responsibly operate in both a functional and regulatory way as well as socially

4.1 Ethical Concerns Around AI Bias and Content Moderation

As TikTok's own AI-driven content moderation and recommendation systems, the concern is in regards to the algorithmic bias. Forbes (2021) highlights that the platform's tendency to downrank or shadow ban underrepresented communities, such as LGBT and political activists. Mixing in human moderation adds inconsistency in the application of the guidelines across regions, and together with the AI models, they may misclassify satire or protest content as harmful.

In reaction to these issues, TikTok has put in an appeals process, published transparency advancements, and vowed to prepare its algorithms with extra inclusive information. Based on this effort, they want to have appropriate engagement optimization balanced with ethical content governance (TikTok Transparency Center, 2023).

4.2 Regulatory Compliance Across Jurisdictions

TikTok works in areas with very different digital regulations. National security concerns in the United States by national security investigations and possible legislative threats, have driven TikTok to propose Project Texas, which involves the storage of U.S. user data on Oracle-managed servers with outside oversight (Reuters, 2023). TikTok is obliged to meet GDPR and the Digital Services Act in the
European Union, obliging the algorithm transparency and child protection measures (European Commission, 2023)

India also banned TikTok in 2020 for concerns over national security, Australia and Canada have imposed restrictions on government devices, and elsewhere, TikTok was banned. These local infrastructure and policy alignment are summarised in table 3 and shown visually in Figure 5 on TikTok's compliance strategies.

Table 3 TikTok Regulatory Landscape by Region (Source: Compiled from BBC, Reuters, and TechCrunch)

Region	Regulation Focus	TikTok Response

United States	Data privacy, national security	Project Texas, congressional testimonies
European Union	GDPR, DSA, algorithm transparency	Local data centers, age controls, and transparency tools
India	National security, content moderation	Full ban (2020)
Australia	Government use restrictions, data handling	Public sector device bans
Canada	Privacy and misinformation policies	Risk assessments, feature limitations

Figure 5 TikTok's Global Compliance Efforts (Source: TikTok Transparency Center, 2023)

4.3 Sustainability and Environmental Responsibility

It is possible for TikTok's scale to also mean high energy consumption in its data centers and real time AI infrastructure. According to ByteDance, parent company, it will become carbon neutral, including the use of renewable energy and green coding practices related to server optimization (GreenBiz, 2023). Some of the social sustainability includes screen time reminders, mental health campaigne, and digital literacy programs. While TikTok has yet to publish a full ESG (Environmental, Social, and Governance) report, or commit to third party carbon audits like Google and Apple, for instance, they have not found much shame in apparently letting life go on business as usual.

4.4 Professional Conduct and Global Operations

Thus, TikTok has been professionally stepping to standardize internal ethics policies, to launch diversity and inclusion programs, and to create global escalation procedures to handle any legal issue globally. Internally, ByteDance has broadened international leadership to drain the centralization of power in China to indicate TikTok as a neutral global platform instead of a Chinese product export (Financial Times, 2023).

Essentially, TikTok's future will be based on how the app can continue to innovate but also address concerns of ethical, legal, and regulatory issues. It must grow aided by its own AI in full accordance with the global standards of transparency, privacy, and sustainability to continue being a trusted platform that both entertains and responsibly molds the developing digital economy.

4.Twitter (x)

1. Unique aspects of the selected business

Since twitter is a source of real time news. Due to the limit of "tweet" size being limited to 280 characters, information about current events would be short and direct. Leaving out unnecessary information or tangents. Adding a subscription, is a way for X to earn more revenue. By implementing 3 different levels, each with their own benefits which allows X to earn monthly revenue. By adding an option to pay for better privileges with monthly payment allows the user to gain special privileges. Switch to this dynamic shows a bit more of a financial driven decision making.

X has a unique property when it comes to how people interact with one another. People would make comments and be quite recognisable. This allows for more user conversation that can grow. With the ability to tag people to join in the conversations it more of a discussion platform like Reddit than a place to post what you're doing. However, being a discussion-based platform can cause its unique problems. With people believing strongly with opposing points can create a hostile disagreement.

After Elon's take over some users ended up leaving the platform, but it remained to be a popular platform for a lot of people to use. Currently twitter has around 611 million users monthly. The gender split being 68.1% to 31. The age ranges that are on twitter range from 13 to over 50 with majority being in the 25-34 age range being 36.6% of all users. With the majority of users being in America with over 100 million users.

Age Groups	**Percentage Of Users (%)**
13-17	**2.4%**
18-24	**34.2%**
25-34	**36.6%**
35-39	**19.7%**
50+	**7%**

X has started implementing blockchain. This improves security and authentications. By having users to verify their identity. This hopefully reduces fake accounts and bots. It also allows facilitation of automated agreements such as subscriptions and direct transfers between users. This opens up opportunities for such as e-commerce,

advertising and other collaborations. Due to improved recordability of tweets, it is much easier enforce accountability of what type of information is shared online.

2. Business Models and Related Software/IT Technologies

A part of the revenue that X receives is thanks to the subscriptions previously mentioned. They have three tears. The first level being “Basic” allows users gin features such as post editing, longer posts and prioritization for replies. This is 3 pounds a month. The next level is “premium” which is 8 pounds a month. Aswell as previous perks, it also provides reduced adds, ID verification and better prioritization. Finally, for 16 pounds a month is “premium +” gives higher prioritisation and removed adds completely. Furthermore, addition revue is through promoting certain adds. These are paid accounts, trends and tweets which allows advertisement on the app in exchange for twitter getting more money. They use blockchain for accountability and security between payments. X also uses paid advertisement. This is for companies to pay X to have rights to send advertisement towards other people. This increase revenue since multiple companies and people can pay for this.

Twitter uses machine learning (ML) to record and analyse user’s liked tweets, posts and who people follow and engage with. This is then used to predict what the user would like to witness and send that type of content to the appropriate user. AI is also used for moderation. It is used to automatically detect harmful language, for example hate speech and threats. After the AI analysis these tweets it can sometimes remove the tweet themselves or flag it for a human to proofread and decide for themselves.

X’s business strategy has been focusing more on marketing, paid ads and subscriptions. Some more specific strategies are native contents, engaging with the users for “playful banter”. Using paid advertisement for higher revenue and to improve revenue even further, implementing monthly and yearly subscriptions. These strategies helped to improve awareness for X and improved customer engagement to a degree. Having an account to be made free allows other businesses to have an option to use X to expand themselves effectively increasing the number of users. Due to what type of content is put on X, it could create a negative appearance for X as people could become concerned on what is posted online. Furthermore, criticise can affect product reputation and even effect the reputation of X. On top of that, the algorithm of X is relied on to produce appropriate content and maintaining the algorithm can drain resources. If it's not properly coded it could cause issues and complaints.

3. Business and Software/IT Alignment Strategies

Using AI to create an algorithm for properly provide the appropriate contents for the users is important. Therefore, a fair amount of resources would be needed and dedicated to provide a sufficient level of quality. Being a social platform, monitoring what is posted online is important. Making sure bots and false accounts are removed efficiently. This means strategies such as using ai to monitor all messages and pinged admins for suspicious activity. Maintaining a level of security is important. For a social media platform, being able to handle a lot of data being sent. Making sure its resistant to DDOS should be a high priority. Unfortunately, it fell victim to a DDOS attack back

in March. Dues to this, IT should put more resources into improving the ability it's servers can manage data.

Using blockchain, security is enhanced by a combination of multiple features such as cryptography which uses encryption and digital signals to keep track and protect data and transactions. Decentralised databases (a characteristic of block chain) allow multiple nodes to exist across then network. If a single node is disabled, the network can still operate along the other nodes instead making it more resilient against DDOS attacks. This also improves the tracking of how data and money is transacted. This is because when contents are added to the blockchain, it cannot be altered. This means there is definitive proof of what data is sent from where. It also allows clean transparency since many blockchains are public ledgers, allowing anyone to see what transactions has occurred.

Moderating the content comes with its own challenges. The most predictable issue is the large volume of users that post all sorts of stuff. Moderating a large amount of data is difficult within itself. The second issue is dealing with the balance between allowing people to speak their mind and removing harmful content such as hateful speech and propaganda. Decided the difference can be a challengingly thin line. Furthermore, making sure everyone keeps to the policies and enforcing these policies over millions of users come with its own properties.

Some methods used to help manage moderations is AI. Machine learning is used to search for key phrases or words to ping for a human user to double check. This helps manage the large number of messages sent since AI can scan through sent data much faster than any human. To help enforce that people are following policies, X moderators will be able to remove messages, issue warning or even suspend accounts. Worse case scenario remove accounts/ suspend indefinitely.

4. Professional, Legal, Ethical, and Sustainability Considerations

Misinformation is a concerning issue for X. There is a concern that there might be Russian bots posting disinformation campaigns since the Ukraine invasion. This was during a European election. This wouldn't be the first time Twitter failed to stop and reduce these issues, since back in 2016, there were reports of similar bots interfering with the news spread online during the US elections as well. This shows a concerning lack of moderation and care from the owners and moderators of X. This shows a lack of professional management, not being able to find and remove these bots till after they have done their damage.

X, being a place for people to speak their minds, can cause many ethical dilemmas. One being how much freedom of speech they have. Due to the policies and managing to remove all hate speech and inappropriate content, it becomes a dangerous thing if people start accusing X of removing what they post as a way to silence them. This could, in turn, affect X with a bad reputation if it gets enough traction. Furthermore, making sure bots are effectively managed and removed if importing them otherwise, it could further ruin X's reputation to the public eye.

To improve the sustainability of X and future management, machine learning is used to help manage an appropriate number of resources to the correct locations. Since it can do calculations much faster than a human can, it can calculate how much data is needed for certain parts of the network to run appropriately. Improving on current systems, like the current AI for managing and identifying inappropriate content. This will allow more fairer experience for users and improve user satisfaction, allowing X to become more sustainable as the years go by.

X can be a considerable legal issue since it can allow for propaganda to grow and spread across the networks. This can be quite problematic and cause a lot of pressure of X, even causing it to be banned from certain countries if it's not actively managed and removed. This applies high pressure and X as they must find and removed these things; however, with things such as bots, there can be cases where more propaganda is added that X cannot remove.

5.Instagram

1. Unique aspects of Instagram

Instagram, which is owned by Meta platforms, is currently the third largest social media platform. Since it was introduced in 2010 it has established itself as the most popular pictorial-sharing mobile app (Hu, Manikonda & Kambhampati, 2014). Instagram has allowed users to connect globally by sharing content through pictures and reels. Businesses have also benefited from Instagram's features as it allows them to promote content and leverage targeted advertising increasing their audience.

One of the main features of Instagram is its role in the influencer economy. From global celebrities to micro-influencers, Instagram boasts a wide range of influencers who can build communities and promote brands organically. Celebrity endorsement greatly increases electronic word-of-mouth (eWOM), which in turn increases consumer trust and product awareness. By leveraging influencer credibility and emotive messaging, these promotions facilitate the transformation of such unfamiliar products into familiar brands (Kutthakaphan & Chokesamritpol, 2013). Instagram has thereby emerged as an effective marketing tool for companies that seek to reach specific consumers.

The value of Instagram has increased even more through its development as a commerce platform. Buyers can easily go from discovering to buying through instruments like Instagram Shopping, product tagging, and checkout. These capabilities increase customer loyalty, product engagement, and brand awareness (Herzallah, Muñoz-Leiva & Liébana-Cabanillas, 2022). Through surfacing products in posts and Stories, sellers can encourage conversion and remove friction.

The second key development is the launch of Instagram Reels as a competitor to TikTok. Reels are now a key part of reach and engagement optimization as they enable the creation of short-form videos (Sisova, 2023). This development is in line with the user's desire for authentic, in-the-moment experiences and reflects wider trends for video-first content strategies. Stories provide ephemeral, real-time content formats that reflect this trend (Sofia, 2024).

Despite the popularity of Instagram, there are several challenges. Foremost among them is competition from TikTok, which has soared in popularity with short videos, particularly with younger audiences (Islam 2025). Then there are algorithm changes that appear to impact the visibility of content and engagement metrics, and this makes it harder for creators and business accounts to experience consistent performance. There is growing concern about mental health effects of social media usage, and this puts the onus on the platform to promote wellness more proactively (Halsall, 2025). Instagram could have soared due to the 2025 TikTok ban in America, but after 2 days it had gotten unbanned, so users went back to TikTok rather than watching Instagram reels.(Aurelio Insisa 2025)

Instagram holds significant growth potential with the introduction of more ecommerce features such as live shopping, personalized feed, and frictionless checkout. AR filters create immersive shopping experiences, and AI-powered content discovery and creator monetization features can be used to improve engagement. Strengthening community-building features will allow businesses and creators to form deeper connections. Moreover, the Meta integration also offers cross-platform synergy, wherein Stories and Reels on Instagram and Facebook can be posted by users, becoming more visible and securing a competitive advantage in social media.

2. Business Models and Related Software/IT Technologies

Instagram's business model is ad-driven, and it has turned the platform into one of the highest-grossing assets for Meta. Brands pay for sponsored posts, Stories, and Reels to reach individuals based on Instagram's sophisticated ad infrastructure. It also makes money from shopping integrations that enable brands to tag products in posts and Stories, drive back to online stores, and provide checkout within the app (Garg, A., 2025). Equally significant are influencer collaborations, in which brands partner with creators to promote products organically, frequently pairing paid advertising with influencer-generated content. ((Kutthakaphan & Chokesamritpol, 2013)

Instagram is based on sophisticated software and information technology to provide an intuitive, personalized experience. Artificial Intelligence (AI) is at its core, particularly in content curation, with algorithms individualizing users' feeds, explore pages, and suggesting Reels according to user behaviour and interests. AI also drives content moderation, automatically detecting and filtering out objectionable or harmful content. One more key technology that is being used to enable interactive filters, virtual

try-ons, and immersive shopping is Augmented Reality (AR), making the platform more interactive for consumers and valuable for brands. In addition, machine learning is applied in ad targeting, user recommendation, and fraud detection. (Stormlikes, 2021).

They have a profound impact on Instagram's business model, corporate structure, and product development. The company is extremely data-driven, constantly tracking and analysing users' behaviour to push engagement and ad effectiveness. AI and machine learning allow Instagram to build its algorithms at a rapid pace, keeping up with market dynamics and users' interests. Product development is usually aimed at driving engagement, time spent on the app, and better monetization prospects for Instagram users and itself. Due to this, Instagram has teams dedicated to AI research, AR development, security, user experience, and business partnerships. Instagram algorithms are known for creating a bubble that becomes so tailored to an individual's preferences that they inadvertently isolate the user from diverse content and perspectives.(Kumar, V , 2024)

Instagrams focus also revolves around empowering creators and businesses, introducing features such as Instagram Insights, Reels rewards, and professional dashboards that enable them to monitor performance, monetize content, and gain presence. Instagram's push for incessant innovation with AI, AR, and e-commerce integrations continues to place it at the top of the rapidly evolving realm of social media.

3. Business and software/IT alignment strategies with a focus on the changing role of new technologies.

Social media Platforms like Instagram allow organizations to reach global markets due to their millions of users worldwide (Green 2018). To support this, Instagram has enhanced its platform by investing in new Technology to align with its business objectives such as increasing user engagement and staying competitive against other social media platforms.

The Meta states that by using more advanced machine learning models, like Two Towers neural networks, they've been able to make the explore recommendation system even more scalable and flexible (Meta Platforms, 2023). Machine learning technology allows the execution of complex processes across their service such as personalizing users feeds to only show content they are interested in. By doing this it keeps users engaged and, on the app longer which supports Instagram's main business objectives of maximizing advertising revenue.

Instagram has also begun using generated AI technologies, such as AI generated captions on videos. This innovation makes Instagram more accessible to a larger audience including people with hearing impairments. This improves the overall user experience and helps businesses reach a more diverse audience. Businesses also benefit from this as they no longer will need to spend time creating multiple versions of their videos to meet different accessibility needs.

As new software and IT technologies emerge Instagram continues to evolve its strategies to remain competitive. For example, as TikTok rose in popularity mostly among a younger audience, Instagram developed reels to match TikTok and keep their users engaged. By introducing Reels Instagram has managed to stay relevant and has provided new ways for user engagement and more advertisement opportunities. This shows Instagram's ability to adapt its technology strategy to meet market demands, reinforcing its position as a competitive platform for both users and advertisers.

However, as Instagram introduces these new technologies some issues may arise such as trust issues with users and lack of data privacy. Therefore, Instagram must balance innovation with ethical considerations.

5.Analysis and Comparisons

1. Presenting Logical Insights

This section makes a systematic attempt to compare the four giant social media networks, i.e., Snapchat, TikTok, Instagram, and Twitter (X), based on their very own defining characteristics, business models, and key enabling technologies. The analysis underlines how these differences have dictated the strategy around each company, their market position, and how the company sustains itself in the marketplace.

2. Unique Aspects and Characteristics

Platform	**Unique Aspects/Features**	**Business Model (s)**	**Key Technologies/IT Stack**
Snapchat	Ephemeral content, , Gen Z focus, AR lenses camera-first interface	Advertising, Subscription (Snapchat+)	Cloud (GCP, AWS), AI/ML, AR, Mobile
		Minor hardware (Spectacles)	Micro-services, NoSQL, Real-time comms
TikTok	Short-form video, AI-powered recommendation, viral	Advertising, Creator Fund, E-commerce	AI/ML, Video streaming, Cloud, Mobile

	trends, creator economy		
Instagram	Visual-first, influencer marketing, integrated shopping, Stories/Reels	Advertising, Shopping, Creator tools	Meta cloud, AI/ML, AR, Mobile
Twitter (X)	Real-time news, open discourse, trending topics, monetization for creators, rebranding	Advertising, Subscription (Blue)	Cloud, AI/ML, Real-time data, APIs
		Creator monetization (Tips, Spaces)	

3. Comparison by Key Strategic Criteria

Company	**Scalability**	**Security**	**Cost-Efficiency**	**Innovation (AI/AR)**	**User Demographics**	**Monetization Diversity**
Snapchat	High	Medium	Medium	High(AR)	Gen Z, Young	Medium
TikTok	High	Medium	High	High(AI)	Gen Z, Millenial	High
Instagram	High	High	High	Medium(AI/AR)	Broad, 18-34	High
Twitter (X)	High	Medium	Medium	Medium(AI)	Broad, Adult	Medium

4. Business Model and Technology Alignment

Company	Primary Revenue	Secondary Revenue	Key Tech Advantage	Strategic Focus
Snapchat	Ads	Subscriptions, Hardware	AR Lenses, Cloud Scalability	Youth engagement, AR commerce
TikTok	Ads	E-commerce, Creator Fund	AI recommendations, Video	Viral content, Creator tools
Instagram	Ads	Shopping, Creator tools	Integration with Meta, AR	Influencer, Shopping
Twitter (X)	Ads	Subscription, Creator tips	Real-time data, Open API	News, Monetization

5. Analysis: Strategic Impact and Market Position

Snapchat

- **Strengths:** Strong AR innovation with a youth focus ensures the highest engagement of Gen Z. By pressing the ephemeral angle, it provides genuine sharing among young workshops and emphasizes privacy (Bayer et al., 2016).
- **Weaknesses:** Monetization is less diversified compared to rivals; struggles to broaden demographic reach.
- **Strategic Impact:** Heavy investment in AR and cloud ensures the scalability required to maintain relevance in the youth market, while innovation is kept under pressure through competitive imitation, i.e., Instagram Stories.

TikTok

- **Strengths:** The market leader in AI-powered content curation for viral trends and user retention. Monetization is swiftly diversifying in e-commerce and creator funds.
- **Weaknesses:** Under regulatory scrutiny and security concerns surfacing in Western markets.
- **Strategic Impact:** This is the very essence of how AI/ML enables the platform to scale rapidly and engage massively; however, sustainability depends on complying with global regulatory challenges (Sun et al., 2023).

Instagram

- **Strengths:** Strong demographic appeal, solid integration of shopping and influencer marketing under Meta security.
- **Weaknesses:** Innovations seem to follow competitors (such as Reels after TikTok); hence, there might be an overreliance on advertising.
- **Strategic Impact:** Strengthened by integration within Meta's ecosystem and a commercial emphasis, the platform finds that offensive by saturation and competition from TikTok to court younger users.

Twitter (X)

- **Strengths:** Real-time conversations and news dissemination, open software architecture, and recent focus on monetization for the creators.
- **Weaknesses:** Monetization is less robust; threats to content security still exist.
- **Strategic Impact:** The rebranding and monetization opportunities pursue a rejuvenation of growth, but platform sustainability depends on regulatory concerns affecting advertisers and supporters of unregulated discourse (Kwak et al., 2023).

6. Conclusion

The comparative analysis conducted shows that all four platforms feature high scalability and are technologically advanced, but they target different strategic focuses, user bases, and drivers of innovation. Snapchat and TikTok both lead in AR and AI innovation; Instagram is best at commerce and working with influencers; and Twitter (X) pushes real-time discourse and creator monetization. These differences drive their competitive strategies and market sustainability, supporting both academic and industry points of view.

4. Professional, Legal, Ethical, and Sustainability principles

Professional principles

As Instagram is owned by Meta, the company follows many professional principles. They aim to build a community and give people a voice while keeping everyone safe and protecting their privacy. Instagram represents these principles in their policies by following all privacy laws and respecting users' rights.

Legal principles

Instagram aims to make sure that the content people see is authentic to not allow people to use their services to misrepresent themselves or to bully others. Instagram follows law enforcement guidelines for privacy therefore they allow the user to choose how and when they would like to share their data and ensure that their data is safely secured (about.meta.com, n.d.). Instagram mentions that they may also share information about misuse or harmful content with other Meta Companies or law enforcement and that they have teams and systems that work to combat abuse and breaches of our Terms and policies, as well as harmful and deceptive behaviors.

Ethical principles

Instagram aims to protect young individuals who are exposed to social media and are vulnerable. Instagram uses AI to detect cyberbullying and remove any harmful comments or posts. Instagram also offers features which allow the choice of filtering comments, reporting and keeping likes hidden. These features help keep users in control of their accounts and allow them to protect themselves. These principles make

users loyal and confident, building trust. They also ensure data privacy. Instagram ensures that all users can control how much personal data they want to be shared.

Sustainable principles

Instagram promotes sustainability through its platform by suggesting various sustainability focused accounts to users. Meta has said that they are committed to achieving net zero greenhouse gas emissions across Meta's global operations and be 100% supported by renewable energy (meta platforms, 2015). Instagram is also focused on promoting socially sustainable principles such as respecting human rights and investing in organizations that promote social and racial justice.

Conclusion

Briefly, these social media software business segment, represented with Snapchat, TikTok, Twitter (X), and Instagram, have rapidly become a support pillar of global communication, entertainment, and commerce. These firms, while sharing something in common with one another as being digitally focused and user-centric, distinguish themselves on the basis of unique technical approaches, business models, and user engagement strategies. TikTok has revolutionized content watching by its AI-based recommendation algorithm, prioritizing video engagement over social networks, enabling even new users to achieve virality thus, making TikTok an innovation leader in algorithmic personalization and social shopping, especially with the growing success of TikTok Shop. Snapchat, on the other hand, remains dominant in the realm of ephemeral content and AR, with offerings like Lenses and Spotlight, supported by cloud infrastructure and enormous investments in AR research and development; it is especially good at engaging with the Gen Z crowd. Meta-owned Instagram is resilient by integrating AR and machine learning into its DNA to drive influencer-driven marketing and e-commerce features such as Instagram Shopping and Reels, while keeping tight to visual-centric engagement and brand building. On the other hand, Twitter (X) stands out as a live discussion platform with focus on short form commenting, subscription monetization, and blockchain identity verification for reducing bots and boosting content accountability, while incurring moderation and user trust under its interim leadership. Business-wise, all platforms survive mainly on ad revenue, but are diversifying into subscription, ecommerce, and virtual gifting models to future-proof their business. Technologically, the players leverage cloud computing, mobile app first strategies, and advanced AI/ML techniques to improve user experiences, automate moderation, and optimize ad targeting Snapchat and Instagram also lead the way in rolling out AR, with Twitter testing blockchain to make it transparent and resilient. Each platform is morally and legally challenged on issues

of data privacy, youths' protection, content moderation, and algorithmic transparency, with compliance needs under regulations like GDPR, COPPA, the DSA, and local data regulations necessitating companies to incur costs in safety elements, transparency, and governance. Professionalism and users' trust are increasingly tied to the quality with which these platforms manage challenges like misinformation, online abuse, and digital well-being, particularly among youths. On the sustainability front, initiatives range from Meta's net-zero operations to TikTok's mental well-being tools and Snap's DEI initiatives, but third-party auditing repeated over time is not common. The future of the space will be the further convergence of AI, AR, and commerce, greater personalization tempered by the transparency of algorithms, and expanding monetization opportunities that empower creators without compromising ethics. Collectively, these websites impact not only how people connect and access information but also how businesses promote, governments regulate, and cultures communicate, making the market for social media software a significant, competitive, and constantly changing ground with far-reaching implications for global digital culture as well as the economy.

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